JOEL LAL

Computer Science & Artificial Intelligence Undergraduate | Desktop Software & AI Prototyping

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PROFESSIONAL SUMMARY

Third-year Computer Science and Artificial Intelligence undergraduate at Murdoch University Dubai with practical engineering experience in Python desktop application architecture, multiprocessing engines, and multi-agent AI systems. Co-author of accepted and presented research at ITT 2026 (proceedings in progress). Experienced in production-ready open-source GUI software and live technical staging at FX Designz; seeking flexible technical internships or junior developer roles.

EDUCATION

Murdoch University Dubai | *BSc in Computer Science & Artificial Intelligence* — **Coursework Completion: April 2027** *(Dubai, UAE)*

- **Curriculum & Timeline:** Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Database Systems, Machine Learning Foundations, Object-Oriented Programming, Systems Analysis & Software Design. Degree conferral scheduled for mid-2027.

GEMS Millennium School | *Senior Secondary (CBSE) | Computer Science & Multimedia* — **Completed: 2023** *(Sharjah, UAE)*

- **Academic Focus:** Completed senior secondary curriculum specializing in Computer Science and Multimedia technologies, establishing core foundations in algorithmic problem solving, software logic, and digital media production.

RESEARCH & CONFERENCE PUBLICATIONS

Adaptive Multi-Agent Framework for Personalised Learning Support | *Co-Author & Presenter | Multi-Agent LLMs* — **Presented: Oct 2026 (Publication In Progress)** *(HCT Dubai)*

- Co-authored and successfully presented peer-reviewed research at the International Conference on Information Technology Trends (ITT 2026) hosted by Higher Colleges of Technology (HCT Dubai); official proceedings in publication process.
- Co-engineered an efficient multi-agent architecture coordinating specialized LLMs with a deterministic decision engine for personalized learning support, reducing token usage by 735.6 tokens and latency by 1.88s per interaction.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Java, Assembly, JavaScript, SQL, MicroPython, HTML/CSS.
- **Frameworks & Engines:** CustomTkinter, TkinterDnD2, Pillow, Chromaprint / fpcalc, Mutagen, OpenCV, PyTorch, Multi-Agent LLMs.
- **Tools & Architecture:** Python Multiprocessing Pool, Worker Threading, Inno Setup, PyInstaller, Git, SQLite, Linux/CLI, REST APIs.
- **Operations & Systems:** Desktop GUI Design, Packaging Standalone Binaries (.exe), Live Audio/Visual Systems, Hardware Troubleshooting.

TECHNICAL PROJECTS & OPEN SOURCE

FXsonic – Desktop Audio Tagging & Acoustic Fingerprinting Suite | *Python 3.13, Chromaprint, AcoustID, SQLite | FXcodez* — **2025 – Present** *(github.com/FXcodez/FXsonic)*

- Engineered a standalone Windows desktop GUI app (.exe / installer) that scans multi-format audio files (.mp3, .flac, .m4a, .ogg, .wav, .aiff) and generates acoustic fingerprints using bundled Chromaprint fpcalc.

- Automated metadata resolution and ID3v2.4 tagging against AcoustID/MusicBrainz APIs, featuring persistent SQLite fingerprint caching (cache.db) to skip unmodified files, cover art downloading, real-time table filtering, title cleaning, and duplicate consolidation.

FXbatch – Multiprocessing Image Optimization & Icon Converter | *Python, CustomTkinter, Multiprocessing, Pillow | FXcodez* **2025 – Present** *(github.com/FXcodez/FXbatch)*

- Developed a high-performance desktop application (CustomTkinter GUI) that distributes batch image resizing, format conversion (JPEG, PNG, WebP, AVIF, ICO), and multi-resolution Windows icon generation across all CPU cores via Python's multiprocessing pool.
- Implemented in-memory size and compression telemetry previews via Pillow BytesIO buffers, non-blocking 20ms UI polling for responsive processing of 1,000+ files (~48 img/sec), folder collision policies, and dynamic token naming.

TECHNICAL PROJECTS & OPEN SOURCE (CONTINUED)

Breachpoint – Cybersecurity CTF Training Platform & Sandbox | *Python, Web Sandbox, AI Companion* **2026 – Present** *(Ongoing Project)*

- Architecting an interactive CTF outreach platform featuring simulated webmail sandboxes for social engineering analysis, payload detection, and defensive literacy.
- Prototyping 'BreachBot', an embedded AI companion designed to provide contextual hints, explain vulnerability vectors, and scaffold participant learning.

Hearth AI – Intelligent Health Monitoring & Telemetry Pipeline | *Python, Machine Learning* **2026** *(Academic Project)*

- Constructed an end-to-end telemetry system integrating synthetic biometric data streams, modular hardware loggers, and predictive models to identify anomaly thresholds.

Space42 AI Job Portal – Candidate Assessment Platform | *Python, Flet, SQLite, Asyncio* **Feb 2026** *(GDG Hackathon)*

- Engineered an AI recruitment portal prototype featuring role-based routing, non-blocking asynchronous resume parsing via worker threads, and a secure upload gateway ('airlock.py') with input sanitization against prompt injection.

HandFlow Engine – Real-Time Vision Gesture Control System | *Python, OpenCV, MediaPipe, CLAHE* **2026 – Present** *(Independent Project)*

- Developing a touchless gesture interface tracking 21 hand landmarks at 30+ FPS, integrating CLAHE illumination enhancement for low-light stability and responsive real-time mouse/gesture emulation.

PROFESSIONAL & OPERATIONAL EXPERIENCE

Technical Logistics & Operations Assistant | *FX Designz | Multimedia Operations* **2023 – Present** *(Dubai / Sharjah, UAE)*

- Directing on-site audiovisual engineering, signal routing, and digital media playback for corporate productions; resolving live hardware/software faults under strict schedules to maintain continuous uptime.
- Interfacing directly with corporate clients and production teams to capture technical staging requirements, manage equipment allocation, and oversee setup logistics.

LANGUAGES & DIGITAL PRESENCE

- **Languages:** English (Fluent), Malayalam (Native / Fluent), Hindi (Fluent).
- **Digital Presence:** Personal GitHub (github.com/Bluwarrior) | Open Source Org (github.com/FXcodez) | LinkedIn (linkedin.com/in/joel-lal-781b242b8) | Live E-Portfolio (bluwarrior.github.io).